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# Fast Ion Battery

"I'm sorry, John. The cleantech thesis has not panned out for us and we're under pressure to limit our exposure in this space. We cannot continue supporting Fast Ion."

John Davidson (HBS '05), a partner at Ware Street Capital (WSC) and a board member at Fast Ion Battery, had just received a phone call from Donna Lerner (HBS '05), a partner at Bluelock Ventures. Bluelock had participated alongside WSC and Franconia Ventures in Fast Ion's $10 million Series A financing in mid- 2008, at a time when every venture capital firm wanted exposure to the burgeoning cleantech sector. Three and a half years later, in December 2011, the venture capital landscape looked very different. Although the pace of investment by VCs had recovered from a sharp dip during the worst financial crisis in seventy-five years, cleantech companies funded in the prior decade had failed to deliver on their promise. Only a few companies that had managed to go public or get acquired had generated large returns. In addition, high profile bankruptcies, such as that of solar cell manufacturer called Solyndra, had tainted the sector, leading most VCs to shift their focus away from funding clean energy ventures.

Lerner's call could not have come at a worse time. Fast Ion was running out of cash and needed another round of financing urgently to continue developing its revolutionary battery. Davidson contemplated whether his firm should make an investment to keep Fast Ion going, and if they did, how he would cover Bluelock's pro-rata share.

## Fast Ion's Promise

The technology underlying Fast Ion was developed by a post-doctoral student at MIT, to address the need for radical innovations in the energy storage landscape. In a landmark TED talk entitled "Innovating to Zero", Bill Gates had put this challenge in perspective by noting that "all the batteries on Earth could currently store less than 10 minutes of the world's electricity needs". In order to really exploit advances in intermittent renewable energies such as solar and wind, therefore, there was a need for big breakthroughs in energy storage technologies that could be deployed at scale. In addition, there was a tremendous need for improvements in battery technologies as electric vehicles became more mainstream.

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Professor Ramana Nanda, Senior Lecturer Robert White, and Research Associate Stephanie Puzio prepared this case with the assistance of Research Associate Sid Misra. This case is not based on a single individual or company but is a composite based on the author's general knowledge and experience. Funding for the development of this case was provided by Harvard Business School. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Although the market for such a technology, if successful, was massive, the initial proof had only been documented in a laboratory setting. However, realizing its potential when he saw it presented at MIT's business plan competition, Davidson decided to help finance its commercialization. Fast Ion Battery was formally incorporated in May 2008. WSC and Bluelock funded a $10 million Series A investment, together with Franconia ventures. Davidson, representing the Series A investors, took a seat on the board (see **Exhibit 1** for excerpts of the Series A term sheet and **Exhibit 2** for the Davidson's projected cap table in his initial investment memo for Fast Ion).

Given the early stage of the technology, Davidson and his fellow-investors had decided to tranche the Series A round, such that the first tranche of $4 million was disbursed in May 2008 at a $ 5 million pre money valuation, with the goal of demonstrating the capabilities of the technology, establishing an IP portfolio, and determining the customer segments and their willingness to pay. The second tranche of $6 million at a 1.5X step-up would be disbursed conditional on meeting the milestones outlined in the financing agreement. The team had made slow but steady progress towards achieving the milestones by December 2009. Although not all the milestones had been met, they had secured a prestigious ARPA-E grant worth $2 million that provided both certification and non-dilutive capital to the company. The Series A investors therefore decided to follow through with the second tranche of $6 million. Looking back to that point, Davidson reflected, "As with many of these tranched investments, the startup met some but not all its milestones. It was not quite right and not quite wrong either. The technology continued to have a lot of promise and the team had begun to develop a strong IP portfolio. Although they had not come close to defining a business model, we were heartened by the validation the competitive ARPA-E grant would provide to future investors and therefore decided it was worth paying to see the next card."

## Investor Backgrounds

Davidson and Lerner had known each other for years. They first met in college through a shared group of mutual friends, but remained nothing more than occasional acquaintances. Years later, their paths crossed again when they took their seats in Section B at HBS as part of the Class of 2005. Over the next two years they became really good friends even climbing Mount Kilimanjaro together. They also had similar career interests in entrepreneurship and Venture Capital (VC). At the end their second year at HBS, both Davidson and Lerner graduated with great jobs lined up at boutique VC funds, WSC and Bluelock, respectively.

They stayed in touch over the years and when the opportunity to invest in Fast Ion presented itself, Davidson decided to reach out to Lerner at Bluelock to complete the syndicate alongside Franconia Ventures. Since Lerner knew that, like all other VCs at the time, Bluelock wanted exposure to cleantech, she decided to take it to her investment committee. They supported the investment and Fast Ion received its Series A funding.

WSC and Bluelock were at two very different points in the life of their funds when they made the initial investment in Fast Ion. WSC had just launched a brand new $100 million fund, had made only one other investment as a syndicate in a consumer technology business (and had an estimated net IRR of less than 5%). Bluelock, on the other hand, was five years into a $500 million dollar fund that had already returned over $1 billion. Additionally, the funds had very different investment philosophies: WSC liked to limit the total amount invested per deal to $10-12 million, whereas Bluelock had more flexibility given their larger fund and was able to investment up to $25 million per deal.



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Analogously, Davidson and Lerner had had somewhat divergent career paths during their time as VC partners. Although both were still junior partners, Lerner already had two 'home-run' investments as a lead partner under her belt, whereas Davidson was still working for his first one and he really hoped Fast Ion would be it.

## Bridge Round

In the months that had passed after the second Series A tranche, Fast Ion failed to identify a clear market segment that would use its technology. The technology and marketing teams were constantly being pulled in different directions due to bad management. By June 2011, Fast Ion had made some progress in developing a product for industrial customers, pivoting away from grid-scale storage towards developing batteries for electric vehicles. Although there was verbal interest from potential customers in this new transportation segment, no commercial agreements could yet be held up as proof of market validation. Realizing the founder's lack of success in focusing the team or advancing the go to market strategy, the board initiated a search for a new CEO in September 2011. However, Fast Ion was running out to money and any new CEO wanted to ensure that the company had sufficient runway to implement the changes that were needed before being willing to formally sign on. Davidson knew that since no new investor would come in at such at time, the company would require a $5 million bridge financing round from the existing investors. He had reached out to the other VCs in the syndicate to discuss the possibility of such a bridge and the terms along which it would be structured (see **Exhibit 3** for a recent email exchange between investors, **Exhibit 4** for the draft terms of the proposed Bridge Round and **Exhibit 5** for the revised timelines and milestones Davidson anticipated given the Bridge round of financing).

## Investment Climate

In 2011, venture capital investors had reduced new capital commitments to startups in the cleantech sector. Fewer than expected cleantech investments had achieved good exits through acquisitions or initial public offerings. Even some that had gone public, such as battery company A123 systems, were struggling with large-scale layoffs as demand for their customers' products had weakened due to competition from lower cost providers. Investors had begun to realize that the long time frames and equity intensity of the cleantech sector were a poor match for many of them (see **Exhibit 6**) and Limited Partners that invested in cleantech funds had become wary of the specific competence of venture capital with respect to the industry. Consequently, the majority of venture investors were forecasting that investment in cleantech startups would decline substantially. On the other hand, investment in software, services and consumer focused internet technologies was forecast to continue increasing as a share of total venture capital deals and dollars, due to advancements in cloud computing and open source technology that had significantly decreased the 'cost of learning' in consumer internet and social media companies (See **Exhibit 7**). The investment needed to learn about the viability of cleantech ventures was larger, leading to lower step ups and more dilution for early stage investors as seen throughout the rounds of A123 Systems (See **Exhibit 8**). Additionally, IPOs in software, services and consumer focused internet technologies had recently outperformed those in cleantech (See **Exhibit 9).**



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## Decision time

Davidson faced a real dilemma. On the one hand, there were signs that Fast Ion was about to turn the corner. The CEO search had yielded two prospective candidates who were extremely well-suited to drive the company forward and it seemed as if the company was gaining traction with developing the technology and more capital efficient business model. In particular, Davidson believed that the technology had the potential to not only dominate in the electric car market, but could ultimately play in the much larger, and as yet untapped, market for grid-scale storage, yielding a half-billion dollar exit (on an investment of $ 75 million) if things turned out well.

On the other hand, Ware Street Capital and the two other investors had already invested $10 million into a company that had not performed up to investors' expectations. Would they be throwing more good money after bad by providing bridge financing? Even if Fast Ion did turn the corner, it would have taken an extra $5 million and a year longer than anticipated to start generating revenue. Would other, later stage investors be willing to provide the significant amounts of capital required to get the company to an exit event given the changing investment climate? If not, how much more capital was Davidson willing to commit to Fast Ion without compromising the balance of WSC's portfolio and feeling the heat from his senior partners? These questions became more pressing following Bluelock's decision not to continue backing Fast Ion.

As a venture capitalist, Davidson had experienced challenging situations before. As he reflected on his dilemma, he thought to himself, "there are a hundred different ways I can lose all my money on an investment, but as a venture capitalist investing in risky startups that is par-for-the-course. It happens 50% of the time in each portfolio. But that is the result when I go after opportunities that could be worth billions of dollars if they are successful, and my job is to see if there is a way to make Fast Ion such a company and to own a large enough share of it to make the risk worthwhile."

Was Fast Ion Battery worth saving?



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**Exhibit 1** Summary of Terms of Series A Preferred Stock Financing for Fast Ion Battery

| | |
|---|---|
| **Investors:** | Ware Street Capital, Franconia Ventures and Bluelock Venture Capital |
| **Type of Security:** | Series A1 and A2 Preferred Stock (the "Preferred Stock"). |
| **Valuation:** | The post money valuation for Series A1 Round of Funding shall be $ 9 million ("Post Money Valuation") based on a $ 4 million capital raise and inclusive of a 20% unallocated post-money employee stock option pool. The series A2 tranche of $6 million will be disbursed conditional on meeting the milestones outlined in [Exhibit 5]. The post money valuation for Series A2 Round of Funding shall be $ 19.5 million. |

**Terms of Series A Preferred Stock**

| | |
|---|---|
| **Dividend Provisions:** | The Preferred Stock will have an annual per share non-cumulative dividend of 8% per annum, payable when and if declared by the Board prior to and in preference to any payment of dividends on Common. |
| **Liquidation Preference:** | In the event of any liquidation, dissolution or winding up of the Company, the holders of Series A Preferred Stock will be entitled to receive, in preference to the holders of Common Stock, an amount equal to the Original Purchase Price plus accrued and unpaid dividends. Thereafter, the remaining assets of the Company will be distributed ratably to the holders of Common Stock. A merger or consolidation (other than one in which stockholders of the Company own a majority by voting power of the outstanding shares of the surviving or acquiring corporation) and a sale, lease, transfer, exclusive license or other disposition of all or substantially all of the assets of the Company will be treated as a liquidation event (a "Deemed Liquidation Event"), thereby triggering payment of the liquidation preference described above unless the holders of at least a majority of the Series A Preferred Stock elect otherwise. |
| **Conversion:** | The holders of Preferred Stock shall have the right to convert any or all shares of Preferred Stock, at the option of the holder, at any time, into shares of Common Stock. One share of Preferred Stock shall initially be convertible into one share of Common Stock (see anti-dilution provisions). |
| **Anti-dilution Provisions:** | The conversion price of the Preferred Stock shall be subject to adjustment according to a broad based weighted-average anti-dilution formula in the event of the sale of the Company's securities at a price less than the current conversion price, subject to standard carve outs. In addition, there will be proportional adjustments for stock splits, stock dividends, recapitalizations and the like. Securities issued pursuant to a Company stock option plan shall not trigger anti-dilution protection. Holders of a majority of the Preferred Stock may elect to waive on behalf of all holders of Preferred Stock the foregoing anti-dilution provisions. |
| **Voting Rights:** | The holders of Preferred Stock will vote together with Common Stock and not as a separate class except as specifically provided herein under "Protective Provisions" or as otherwise required by law. Each share of Preferred Stock shall have a number of votes equal to the number of shares of Common Stock then issuable upon the conversion of such shares of Preferred Stock. |



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**Protective Provisions:**

The consent of holders of a majority of Preferred Stock, voting as a separate class, shall be required to:
a) change the rights and terms of the Preferred Stock;
b) issue any security either senior to or on a parity with the Preferred Stock
c) any items adversely affecting the rights and privileges of the Investors
d) allow a transfer of stock by the Founders
e) declare dividends and redeem or buy back shares
f) incur any material indebtedness
g) allow related party transactions
h) amend any option plan or bylaws of the Corporation
i) change the number of directors
j) allow the liquidation, recapitalization or reorganization of the Company
k) any significant sale of assets of the company, including any sale of Intellectual property
l) sell or merge the Company
m) change in senior management (CEO and CFO) of the Company
n) appoint statutory auditors to the company
o) amendments to Memorandum of Association and Articles of Association of the Company

## Terms of Investor Rights Agreement

**Board of Directors**

The Board of Directors will consist of 3 members, to be elected as follows:
CEO of the Company,
John Davidson, who will represent the investors
One independent outside director who shall be mutually acceptable to Common shareholders and to majority of Preferred Investors.

**Right of First Refusal And Co-Sale Agreement:**

If the Founder(s) receives an offer to sell all or any portion of his/her shares of capital stock in the Company, both the Company (first) and the holders of Preferred Stock (second) shall have the right of first refusal to purchase such shares on the terms of the offer. If the Company and Preferred Shareholders decline this option, the Founder may not accept such an offer unless a similar offer on the same terms is made pro rata for the securities held by the holders of Preferred Stock. This shall not apply to transfer of stock by Founder(s) to their trust or family members.

**Drag Along Right:**

At any time after 60 months from the date of closing, the Investor shall have the right to transfer their Preferred Stock to any other person. In relation to such proposed transfer, the Investors shall be entitled to require the Founder to transfer all or some (to be decided at the sole discretion of the Investor) of the equity securities held by the Founder to such other person at the same price per share and on the same terms and conditions applicable to the Investor.

## Other Matters

**Employee Common Stock Options Vesting**

Unless otherwise approved by the Board, Employee Common Stock Options shall vest based on an ESOP scheme approved by the Board of the Company and the Investors. The Company shall have a repurchase option on unvested shares at cost



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| | |
|---|---|
| **Founders Common Stock Options Vesting** | Unless otherwise approved by the Board, Founders Common Stock Options shall vest as follows: 25% will vest at the end of one year; the remainder shall vest monthly over the following 36 months. The Company shall have the right to repurchase any unvested Common Stock at cost upon termination of the Founder's employment with the Company |
| **ESOP:** | Independent of the amount raised, the company will create an unallocated Employee Stock Option pool available for grant to 20% of the outstanding shares post-closing of the Investment. All equity or option grants to all employees, consultants and directors shall be subject to minimum 4 year vesting and other typical restrictions with a twelve month cliff upon initial hiring. |
| **Closing:** | Expected on or about 15th May 2008. |
| **Fees and Expenses:** | Upon successful completion of the Investment, the Company will pay reasonable fees and expenses of a counsel for the Investors, financial and background due diligence and travel expenses up to a maximum of $ 20,000. |
| **Non-Binding:** | Except for the Confidentiality and Exclusivity provisions, and Fees and Expenses, which are binding agreements among the undersigned, the undersigned acknowledges that this term sheet does not constitute a binding agreement. |
| **Expiration:** | If not accepted by the Company, the offer contained in this term sheet will expire on 30th March, 2008. |
| **No Shop:** | From the signing date hereof until the earlier of (i) the Closing or (ii) written notification by Ware Street Capital that it does not intend to proceed with the financing, the Company and the Founder agree that they shall not solicit, encourage others to solicit, encourage or accept any offers for the purchase or acquisition of any capital stock of the Company, of all or any substantial part of the assets of the Company, or proposals for any merger or consolidation involving the Company, and they shall not negotiate with or enter into any agreement or understanding with any other person with respect to any such transaction. |

Source: Casewriters.



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**Exhibit 2** Projected Capitalization Table for Fast Ion: prepared by Jim Davidson in his investment memo to WSC's partners in March 2008.

| | Series A1 ($ 4 M) June 2008 | | | Series A2 ($ 6 M) September 2009 | | | Series B ($ 25 M) September 2011 | | | Series C ($ 35 M) June 2013 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pre-Money --> | $5,000,000 | | | $13,500,000 | | | $50,000,000 | | | $175,000,000 | | |
| Post-Money --> | $9,000,000 | | | $19,500,000 | | | $75,000,000 | | | $210,000,000 | | |
| | Number of shares | Ownership | Value | Number of shares | Ownership | Value | Number of shares | Ownership | Value | Number of shares | Ownership | Value |
| Common (Founders, Mgmt & option pool) | 5,000,000 | 55.6% | $5,000,000 | 5,000,000 | 38.5% | $7,500,000 | 5,000,000 | 25.6% | $19,230,769 | 5,000,000 | 21.4% | $44,871,795 |
| **Series A1 Investors** | | | | | | | | | | | | |
| Ware Street Capital | 1,500,000 | 16.7% | $1,500,000 | 1,500,000 | 11.5% | $2,250,000 | 1,500,000 | 7.7% | $5,769,231 | 1,500,000 | 6.4% | $13,461,538 |
| Franconia Ventures | 1,500,000 | 16.7% | $1,500,000 | 1,500,000 | 11.5% | $2,250,000 | 1,500,000 | 7.7% | $5,769,231 | 1,500,000 | 6.4% | $13,461,538 |
| Bluelock Ventures | 1,000,000 | 11.1% | $1,000,000 | 1,000,000 | 7.7% | $1,500,000 | 1,000,000 | 5.1% | $3,846,154 | 1,000,000 | 4.3% | $8,974,359 |
| **Series A2 Investors** | | | | | | | | | | | | |
| Ware Street Capital | | | | 1,500,000 | 11.5% | $2,250,000 | 1,500,000 | 7.7% | $5,769,231 | 1,500,000 | 6.4% | $13,461,538 |
| Franconia Ventures | | | | 1,500,000 | 11.5% | $2,250,000 | 1,500,000 | 7.7% | $5,769,231 | 1,500,000 | 6.4% | $13,461,538 |
| Bluelock Ventures | | | | 1,000,000 | 7.7% | $1,500,000 | 1,000,000 | 5.1% | $3,846,154 | 1,000,000 | 4.3% | $8,974,359 |
| **Series B Investors** | | | | | | | | | | | | |
| Ware Street Capital | | | | | | | 975,000 | 5.0% | $3,750,000 | 975,000 | 4.2% | $8,750,000 |
| Franconia Ventures | | | | | | | 975,000 | 5.0% | $3,750,000 | 975,000 | 4.2% | $8,750,000 |
| Bluelock Ventures | | | | | | | 650,000 | 3.3% | $2,500,000 | 650,000 | 2.8% | $5,833,333 |
| Series B Investor | | | | | | | 3,900,000 | 20.0% | $15,000,000 | 3,900,000 | 16.7% | $35,000,000 |
| **Series C Investors** | | | | | | | | | | | | |
| Ware Street Capital | | | | | | | | | | 417,857 | 1.8% | $3,750,000 |
| Franconia Ventures | | | | | | | | | | 417,857 | 1.8% | $3,750,000 |
| Bluelock Ventures | | | | | | | | | | 278,571 | 1.2% | $2,500,000 |
| Series B Investor | | | | | | | | | | 557,143 | 2.4% | $5,000,000 |
| Series C Investor | | | | | | | | | | 2,228,571 | 9.5% | $20,000,000 |
| Total Shares Outstanding | 9,000,000 | 100% | $9,000,000 | 13,000,000 | 100% | $19,500,000 | 19,500,000 | 100% | $75,000,000 | 23,400,000 | 100% | $210,000,000 |

Source: Casewriters.

Note: Ware Street Capital Projected to invest $11.25 million and own ~18.8% at exit, implying a 5.8X return and 45% IRR, conservatively assuming 7 years to exit at a $ 350 million valuation.

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**Exhibit 3** Email Exchange between Fast Ion Investors

**From: Davidson, John**
**Sent: Tuesday, December 6, 2011 4:53 PM**
**To: Mehra, Ajay; Lerner, Donna**
**Subject: Fast Ion**

Donna and Ajay,

I was over at Fast Ion yesterday to get a pretty complete update. In short, while there is some good progress on key fronts we pushed at the last Board meeting, cash is dwindling fast. Given that the CEO search, while very promising with our two lead candidates still in play, is not yet done and the partnership discussions, also while promising, are still in work, the Series B is very unlikely to come together in time, and certainly not at a price or structure that would work well for any of us.

As we agreed, I talked with the Company about pulling together a bridge financing to allow time to get the new CEO on board and to really focus on the market entry strategy and partnership priorities. With the right CEO and a partnership in place, this is a really interesting story and it feels like we could get there in 6 months.

I connected with some of my Partners briefly this morning and there is reasonable support to explore a $5M bridge / Series A extension to get out ~12 months and allow time for this to come together. There is obvious concern about the lack of progress and the overall financing environment, but still enthusiasm for the potential here and if structured correctly, this could be an opportunity to get some more ownership ahead of what could be a significant value inflection point. I believe this can also help to converge the CEO search as both of the candidates are very focused on the runway and having time to get their arms around things before having to get on the road to raise money.

I need to get the Company in next week for a full Partner update, and I'd like to pull together some terms ahead of that, and I suspect you will need the same for your Teams.

Why don't we get on the phone tomorrow or Friday so we can be synced ahead of our Partnership meetings on Monday with the goal of pulling together a set of terms by mid-week next week and be prepared to finance by the end of the month as payroll starts to get tight in early next month.

Kelli will reach out to get a call set up for the three of us…

Thanks,

John

**From: Mehra, Ajay**
**Sent: Tuesday, December 6, 2011 6:18 PM**
**To: Davidson, John; Lerner, Donna**
**Subject: RE: Fast Ion**

John,

Sounds good, I will talk to my Partners in the next couple of days but I provided an update on Monday and I think this plan makes sense based on what we talked about then. Have Kelli coordinate with my admin, I have some time late on Thursday.

Ajay



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**From: Lerner, Donna**
**Sent: Wednesday, December 7, 2011 9:28 AM**
**To: Mehra, Ajay; Davidson, John**
**Subject: RE: Fast Ion**

Guys,

I am pretty concerned about the lack of traction on the CEO search. Are we sure we have the right search firm engaged here? And we did our quarterly portfolio review this week and my guys are feeling pretty squeamish about the Company and the space overall – we are just not seeing the big valuation increases we are seeing in other sectors.

I will talk to a couple of the guys on my Investment Committee to tee this up and get a better feel for things but I think it may be a bit tricky to get support for that large a bridge.

Donna

Source: Casewriters.



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**Exhibit 4** Proposed terms in Bridge Financing for Fast Ion Battery

| Terms | Details |
|---|---|
| Amount of Financing | Minimum of $5M |
| Pro-rata | WSC: $1,875,000<br>Franconia: $1,875,000<br>Bluelock: $1,250,000 |
| Security/Price | A2 Preferred/Flat Round (same price as A2) |
| Terms of Series A2 Extension | Identical to current terms of Series A2 Preferred Stock (Majority vote to change rights, preferences and privileges) |
| Investor Rights Agreement | No change required. (Majority vote to Amend) |
| Oversubscription | In the event an investor purchases more than its pro-rata amount, such investor will be issued Common Stock warrants equal to the amount of the oversubscription divided by the A2 purchase price. The warrants will have an exercise price of $0.50 per share and a 10 year life. |
| Pay to Play: (pending agreement from current Series A investors) | In the event an investor does not participate to its full pro-rata, all its existing Preferred Stock converts to Common Stock at a 10-1 ratio. (e.g. 10 shares of preferred stock would be converted into 1 share of common stock) |

Source: Casewriters.



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**Exhibit 5** Revised Timelines and Milestones after accounting for Bridge Financing

| Round of funding | Series A1 (May 2008) | Series A2 (December 2009) | Bridge - Series A2 Extension (December 2011) | Series B (Projected December 2012) | Series C (Projected June 2014) | Exit - IPO (Projected December 2016) |
|---|---|---|---|---|---|---|
| Money Raised | $ 4 million<br>+ $ 2 million ARPA-E grant | $ 6 million | $ 5 million | $ 25 million | $ 35 million | $ 50 + million |
| Projected months of Runway | 12- 18 months | 18-24 months | 12-15 months | 24 months | N/A Cash flow positive | |
| Projected proof points / Milestones to secure next round of funding | Demonstrate capabilities of technology outside the lab (reproduceability)<br><br>Articulate a board approved IP strategy including a positively completed Freedom-to-Operate (FTO) analysis from an approved third party<br><br>Begin to build-out management team<br><br>Define customer segments | Demonstrate scale up potential of technology<br><br>Define a plan for manufacturing<br><br>Develop business model and generate firm orders from partners in primary market<br><br>Complete building out engineering and sales team | Hire new CEO<br><br>Secure milestones that were missed in Series A2 | Build pilot manufacturing plant with repeatable process for manufacturing at more scale,<br><br>Quality testing and low volume production,<br><br>Start generating revenue but still not profitable. However could be attractive target for potential acquisition by incumbent at ~ 200 M valuation | Build out full manufacturing line capable of $ 100 million of annual revenue<br><br>Leverage the capital of a partner or debt finance (including possibility of venture debt) so that equity investment is limited to $ 35 million<br><br>Focus on operating efficiency so that move towards 15% EBITDA margin and 6% net operating margin | Access public markets to scale further production or sell to incumbent firm<br><br>If achieve milestones can expect exit of $ 300-500 million in December 2016 |
| Actual progress | Secured IP portfolio, won ARPA-E grant for technology, however poor progress in defining market and customer segment | Demonstrated scale up potential of technology; defined capital light manufacturing model and identified attractive customer segment. However no hard agreements or purchase orders; CEO in process of being replaced. Two leading candidates look promising. | | | | |

Source: Casewriters.

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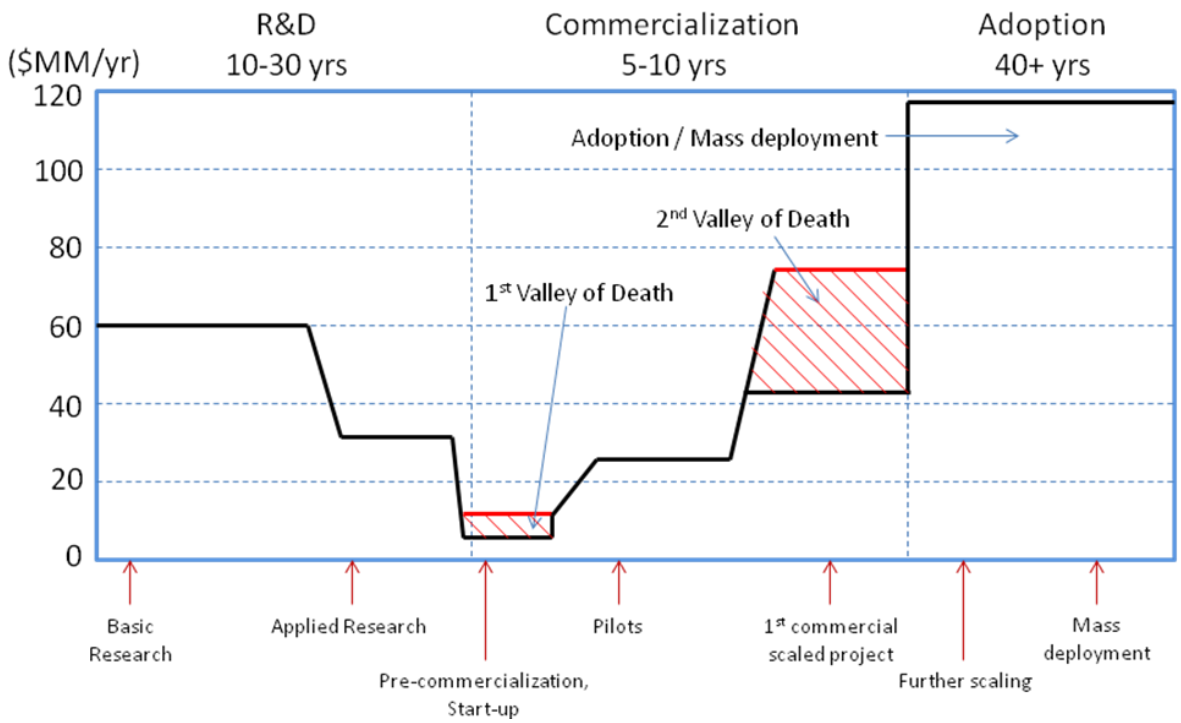
**Exhibit 6** Illustrative Development Cycle for a Representative Clean Energy Technology




Source: Adapted from Mohr Davidow Ventures, 'Capital Profile of a Cleantech Innovation.'

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**Exhibit 7** Venture Capital Financing Landscape: Number of deals and predictions for 2012

| | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 |
|---|---|---|---|---|---|---|---|
| Total Deals | 3,262 | 3,827 | 4,124 | 4,111 | 3,065 | 3,526 | 3,673 |
| Software and Services share of all deals | 42% | 40% | 40% | 41% | 41% | 44% | 45% |
| Computer Hardware and Equipment | 24% | 23% | 20% | 17% | 16% | 13% | 11% |
| Biotech and Medical Devices | 21% | 22% | 22% | 22% | 25% | 24% | 21% |
| Media and Entertainment | 6% | 8% | 9% | 10% | 9% | 9% | 12% |
| Energy and Industrials (including clean tech) | 5% | 6% | 8% | 9% | 8% | 8% | 8% |
| Share of deals that are follow-on funding | | | | | | | |
| Software and Services share of all deals | 65% | 67% | 66% | 67% | 70% | 67% | 65% |
| Computer Hardware and Equipment | 76% | 76% | 76% | 80% | 82% | 79% | 77% |
| Biotech and Medical Devices | 70% | 68% | 71% | 73% | 80% | 77% | 81% |
| Media and Entertainment | 52% | 53% | 55% | 57% | 67% | 59% | 52% |
| Energy and Industrials (including clean tech) | 54% | 49% | 53% | 58% | 69% | 72% | 74% |

Notes:

1. Over 60 percent of venture capitalists predict investment increases in information technology, including Software, Services and Media and Entertainment.
2. On the flip side, 58 percent of industry respondents expect investment decreases in the biopharmaceutical and medical device sectors and 55 percent expect investment levels to decline in clean technology companies.

Source: Adapted from National Venture Capital Association. VC financing data adapted from Q42011 VC investments as reported by MoneyTree. VC Predictions for 2012: http://www.nvca.org/index.php?option=com_content&view=article&id=78&Itemid=102

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**Exhibit 8** Fundraising Rounds and IPO Ownership for A123 Systems

| Round | Date | Investors | Funds Raised | #Shares | Share Price |
|---|---|---|---|---|---|
| A | 2001 | MIT, Sequoia, Northbridge | $8,312,087 | 8,312,087 | $1.00 |
| A-1 | 2002 | | $4,387,500 | 2,925,000 | $1.50 |
| B | 2005 | Yankee, OnPoint, Motorola | $20,017,400 | 9,623,750 | $2.08 |
| C | Jan-06 | Deshpande, Northbridge, QC, GE, Motorola | $30,290,871 | 8,988,389 | $3.37 |
| D | 2007 | Deshpande, Northbridge, QC, GE, Motorola | $69,993,284 | 10,669,708 | $6.56 |
| Common | Jan-08 | | $16,501,454 | 2,285,520 | $7.22 |
| E | May-08 | GE | $102,070,854 | 6,152,553 | $16.59 |
| F | Apr-09 | Deshpande, Northbridge, QC, GE | $99,932,479 | 10,862,226 | $9.20 |
| **Total Pre-IPO** | | | **$351,505,929** | **59,819,223** | |
| **IPO** | | | **$428,315,513** | | **$13.50** |

Note: Founders and Employees (including options) 20,303,624 shares

| Investor | Pre-IPO Ownership | Post-IPO Ownership | Post-IPO # of Shares |
|---|---|---|---|
| Founders | 5.6% | 4.1% | 7,377,878 |
| Other officers and executives | 3.2% | 2.3% | |
| Options | 15.5% | 11.2% | 12,925,746 |
| Northbridge | 10.6% | 7.7% | 8,951,826 |
| General Electric (GE) | 9.9% | 7.3% | 8,482,098 |
| Deshpande | 8.4% | 6.1% | 7,017,629 |
| Qualcomm (QC) | 6.4% | 4.7% | 5,379,526 |
| Motorola | 5.8% | 4.2% | 4,844,914 |
| Other Investors | 34.6% | 25.0% | 28,947,994 |
| IPO | | 27.4% | 31,727,075 |
| **Total** | **100%** | **100%** | **115,654,686** |

Source: Adapted from Herve Lebret, "Start-up: Equity in Start-ups Historical Data," September 2014, accessed at http://www.slideshare.net/lebret/equity-in-305-high-tech-start-ups-september-2014



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**Exhibit 9** IPO Performance for select CleanTech and Software, Services and Consumer-focused Internet Technologies

| Issuer | Ticker | Industry | Founding Year | Offer Year | Offering Price | IPO Market Capitalization | As of November 2011 | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Price | Market Capitalization | % Appreciation |
| Ascent Solar Technologies | ASTI | Semiconductors: Photovoltaic cells | 2005 | 2006 | $5.50 | $29,100,000 | $7.10 | $27,657,000 | 29% |
| Canadian Solar | CSIQ | Semiconductors: Photovoltaic cells | | 2006 | $15.00 | $409,050,000 | $3.36 | $145,100,000 | -78% |
| Clean Energy Fuels Corp | | Energy | 1997 | 2007 | $12.00 | $530,320,932 | $11.91 | $838,400,000 | -1% |
| A123 Systems Inc | AONE | Electrical components: Batteries | 2001 | 2009 | $13.50 | $1,561,338,261 | $2.10 | $242,874,841 | -84% |
| Generac Holdings | GNRC | Electrical equipment: Generators | 1959 | 2010 | $13.00 | $852,524,270 | $24.64 | $1,665,700,000 | 90% |
| SemiLEDs Corp | LEDs | Semiconductors: Light emitting diodes | 2005 | 2010 | $17.00 | $618,744,869 | $3.60 | $98,360,000 | -79% |
| Baidu.com Inc | BIDU | Chinese internet search engine | 2000 | 2005 | $27.00 | $117,311,679 | $146.41 | $51,117,800,000 | 442% |
| Shutterfly, Inc. | SFLY | Internet Retail: photography | 2005 | 2006 | $15.00 | $354,166,920 | $41.67 | $1,444,700,000 | 178% |
| LinkedIn Corp | LNKD | Internet software and services: Social network | 2003 | 2011 | $32.00 | $3,651,538,208 | $87.50 | $8,437,200,000 | 173% |
| Zillow Inc | Z | Internet software and services: Online real estate | 2004 | 2011 | $18.00 | $625,457,354 | $32.63 | $901,800,000 | 81% |
| Zipcar Inc | ZIP | Passenger car rental: Car sharing | 1999 | 2011 | $18.00 | $818,052,156 | $17.86 | $701,900,000 | -1% |
| Pandora Media Inc | P | Internet software and services: Music streaming | 2000 | 2011 | $16.00 | $3,225,252,960 | $9.85 | $1,589,500,000 | -38% |

Source: Capital IQ; Nasdaq.com; Digitallock.com; Jay Ritter and Tim Loughran, "Why Has IPO Underpricing Changed Over Time?" accessed at http://bear.warrington.ufl.edu/ritter/FoundingDates.htm; Casewriter analysis.

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